

Eisha Khan

Threat Hunting and Incident Response — Final Project

THREAT HUNTING · INCIDENT RESPONSE · DIGITAL FORENSICS

PART 1 — ENVIRONMENT DETAILS AND THREAT HUNTING

For Part 1 of the final project, I performed an initial threat hunting and environment reconnaissance exercise to understand the network before beginning an incident investigation. Since no documentation about the environment was provided, I had to identify every system, determine its purpose, and gather information using multiple security tools.

The first step was logging into each of the provided security platforms. These included Splunk Enterprise, Security Onion, Velociraptor, and Arkime. Before beginning any searches, I adjusted the time range in each platform to ensure I was searching the complete dataset rather than only recent events.

• Using Velociraptor

Velociraptor was the primary endpoint forensic tool used during this investigation. I accessed the Host Information page for every endpoint in the environment and documented information including hostname, Fully Qualified Domain Name (FQDN), domain membership, operating system, operating system version, IP address, MAC address, Client ID, domain role, and system architecture.

From this information I identified all nine hosts within the environment:

- Earth
- Mercury
- Venus
- Mars
- Jupiter
- Saturn
- Uranus
- Neptune
- Pluto

I determined which systems were Windows workstations and which were Windows Servers. I also identified that **Earth** functioned as the Primary Domain Controller because of its server role and Active Directory responsibilities.

Enumerating User Accounts

Using the `Windows.Sys.Users` artifact in Velociraptor, I collected local user information from every endpoint. For every host I documented administrator accounts, guest accounts, the SYSTEM account, LOCAL SERVICE, NETWORK SERVICE, additional local users, domain users, and last-login timestamps where available. I also reviewed `Windows.EventLogs.RDPAuth` logs to determine which users had authenticated through Remote Desktop. This allowed me to identify typical user accounts, service accounts, built-in Windows accounts, Remote Desktop users, and recently logged-in users, and to determine which users typically accessed each machine.

Software Enumeration

To determine installed applications, I collected the `Windows.Sys.Programs` artifact. The software inventory contained hundreds of installed applications across the environment. Since the assignment required omitting Windows default software, I analyzed the results to identify non-standard applications — examples included Brave Browser, Brave VPN, Brave Update Services, Elastic Endpoint, Splunk Universal Forwarder, and various third-party utilities. This demonstrated the ability to distinguish between default Windows components and software intentionally installed by administrators.

Service Enumeration

Next, I collected the `Windows.System.Services` artifact, which provided a complete inventory of Windows services running across every system. I analyzed the service list to determine security software, browser update services, endpoint monitoring services, Windows default services, and services unique to certain hosts — examples included SplunkForwarder, Elastic Endpoint, Brave Update Service, Brave VPN Service, Windows Defender services, BitLocker services, Background Intelligent Transfer Service, and Base Filtering Engine. By comparing services across systems, I identified hosts running additional security software or third-party applications.

• Using Splunk

Splunk was used as the centralized log management and SIEM platform. I performed searches across all indexes using queries such as `index=*`, and searched Windows event logs for authentication events, process creation, file creation, registry changes, and Sysmon events. During my investigation I examined successful logons, failed logons, process execution, file creation events, and endpoint activity, and reviewed Sysmon logs to determine whether any suspicious processes or file modifications had occurred.

Investigating File Creation Events

One Sysmon FileCreate event initially appeared suspicious because it involved creation of files containing the phrase `"DO-NOT-TOUCH"`. Upon further investigation, I identified that the process responsible was `elastic-endpoint.exe`, located inside `C:\Program Files\Elastic\Endpoint\`. The files were part of Elastic Endpoint's Anti-Ransomware protection mechanism. Rather than indicating malware, this activity represented normal behavior performed by endpoint protection software. This demonstrated the importance of validating alerts before classifying them as malicious.

• Using Security Onion

Security Onion was used to monitor network activity and IDS alerts. Within Security Onion I examined alerts, network events, Zeek logs, and Suricata detections. I searched for indicators such as suspicious network traffic, unauthorized connections, high-severity IDS alerts, external communications, and authentication activity. I correlated these events with information collected from Splunk and Velociraptor to determine whether alerts represented real threats or normal activity.

- **Using Arkime**

Arkime was used for packet capture analysis. Within Arkime I investigated network sessions, source IP addresses, destination IP addresses, protocols, ports, and session duration. I reviewed network communications between hosts to determine which systems communicated most frequently, whether external connections existed, and whether unusual protocols were present. Arkime provided additional evidence supporting host activity observed within Splunk and Security Onion.

- **Threat Hunting Methodology**

Throughout Part 1 I continuously hunted for potential threats by examining unusual logon activity, suspicious processes, unknown services, unexpected software, file creation events, RDP logons, network traffic, security alerts, and endpoint artifacts. Rather than relying on a single tool, I correlated information across multiple platforms to validate findings. This process demonstrated how endpoint forensics, SIEM analysis, IDS alerts, and packet captures work together during an investigation.

PART 2 — INCIDENT INVESTIGATION

For Part 2, I conducted an independent investigation without guided instructions. The objective was to identify suspicious activity using the tools provided and determine whether a security incident had occurred.

I began by reviewing alerts inside Security Onion to identify any hosts that generated IDS events. Next, I examined Splunk logs for authentication events, Sysmon activity, process creation, and file creation events. Using Velociraptor, I gathered endpoint artifacts including user accounts, installed software, running services, host configuration, and endpoint information, and compared them with SIEM logs to identify inconsistencies or suspicious behavior. I then used Arkime to investigate network communications involving the affected hosts, verifying communication patterns and determining whether suspicious outbound or internal traffic was present.

During the investigation I discovered several security events that required validation. One example involved Sysmon FileCreate events generated by Elastic Endpoint. Initially these events appeared suspicious because they involved files with unusual names. After correlating process information, executable paths, and service data, I determined the activity originated from Elastic Endpoint's ransomware protection features and represented expected security software behavior rather than malware. This demonstrated the importance of validating alerts before escalating them as incidents.

Throughout the investigation I correlated evidence from multiple tools rather than relying on a single alert source. Using endpoint telemetry, centralized logging, intrusion detection alerts, and packet captures allowed me to reconstruct system activity and determine whether observed events represented malicious behavior or legitimate administrative actions. This investigation reinforced the incident response methodology taught throughout the course by combining threat hunting, log analysis, endpoint

forensics, network analysis, and evidence correlation to accurately assess the security posture of the environment.

• Skills Demonstrated

- Threat hunting methodology
- Incident response workflow
- Splunk SIEM log analysis
- Security Onion investigation
- Zeek and Suricata alert analysis
- Arkime packet analysis
- Velociraptor endpoint forensics
- Windows endpoint investigation
- Sysmon event analysis
- User enumeration
- Software inventory analysis
- Windows service analysis
- Authentication investigation
- RDP log analysis
- Host profiling
- IOC identification
- Alert validation
- Cross-platform log correlation
- Network traffic analysis
- Endpoint artifact collection
- Security documentation and reporting

This project provided hands-on experience performing a real-world threat hunting and incident response investigation using enterprise security monitoring and digital forensic tools.